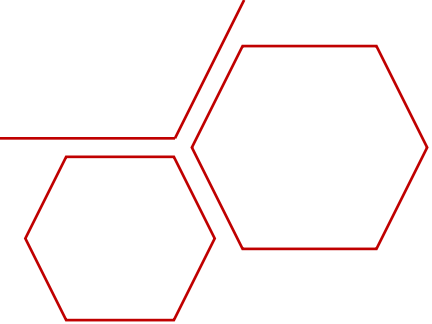




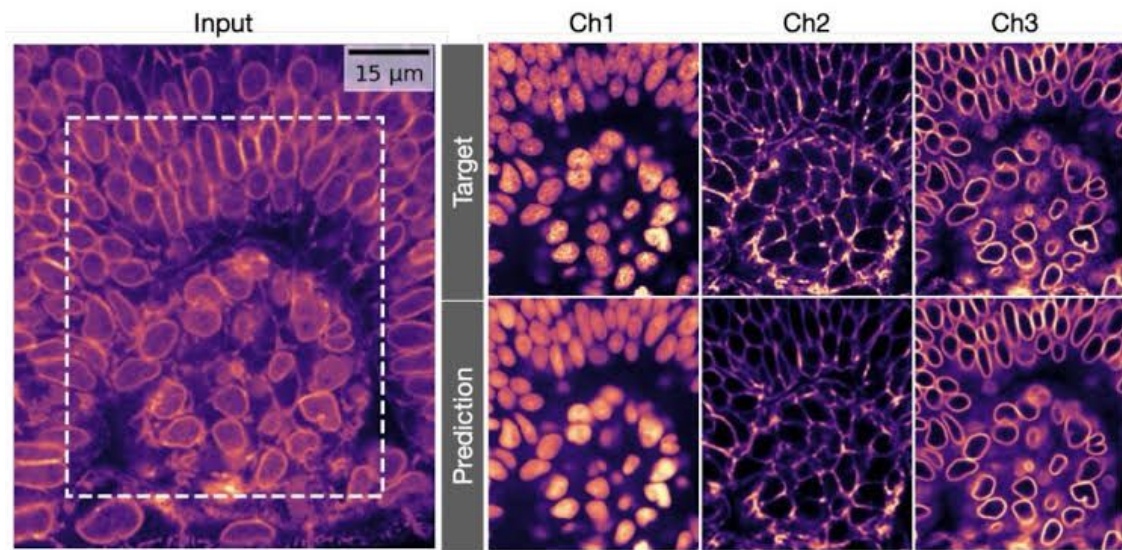
Optimizing Cell Segmentation Software

Jayden Hu

Mentors: Jingxuan Jia, Haofan Sun
Fei Lab

Modern Microscopy Produces Massive Datasets

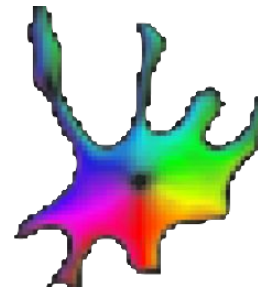
- Cell microscopy is used in biology, medicine, and cancer research
- Current imaging techniques produce thousands of images



Source: Human Technopole

No Single Model Works Best for Every Dataset

- Segmentation model performance varies across each dataset
- **Question: How can researchers find the segmentation model best suited for their dataset?**



Simplifying Segmentation Model Evaluation

Goal: Develop a reproducible benchmarking framework for microscopy segmentation models

Benchmarking
Framework

- Objective evaluation



Segmentation
Models

- Improved accuracy



Quantitative
Results

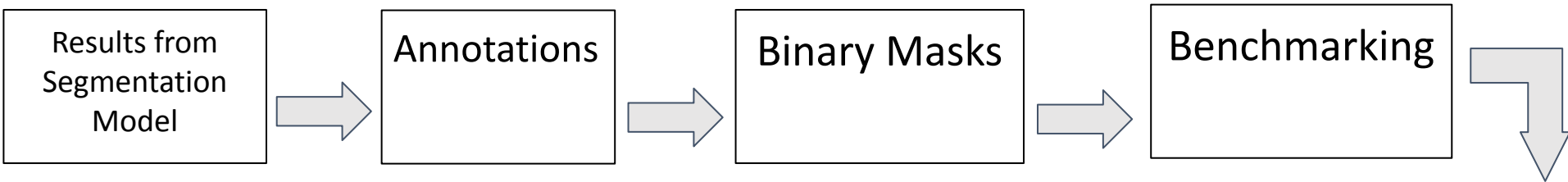
- Reliable measurements



Downstream
Biological Analysis

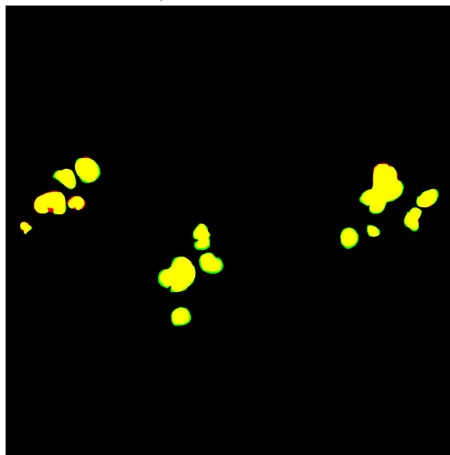
- New breakthroughs

Benchmarking Framework Workflow

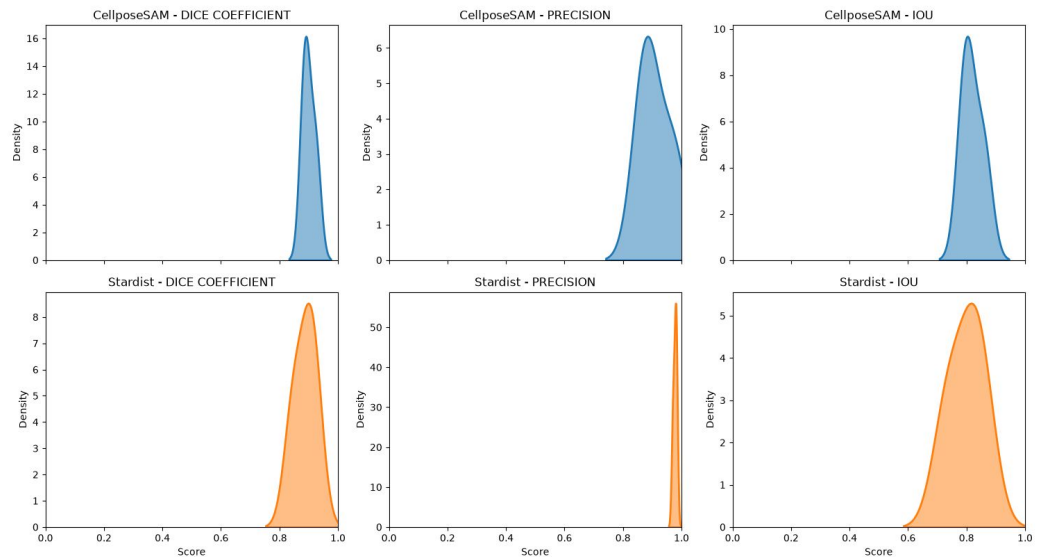


Sample Visualizations

CellposeSAM vs Manual

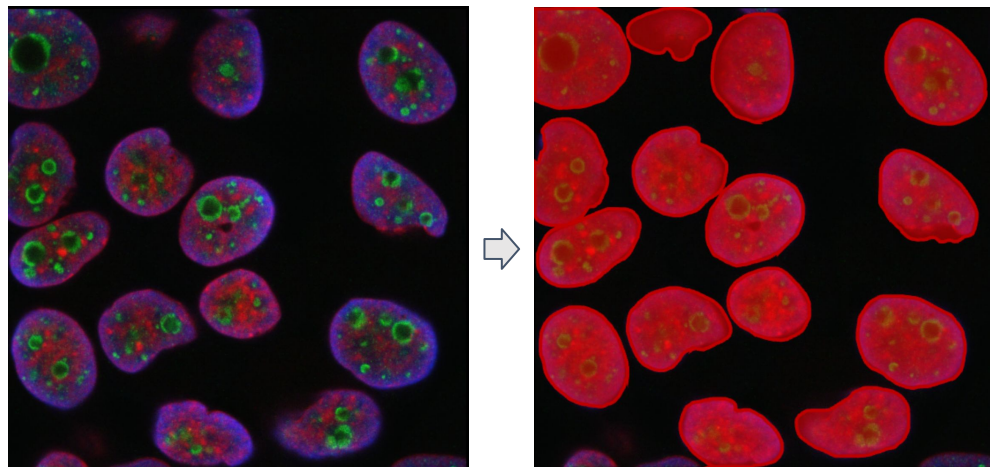


- True Positive (TP)
- False Positive (FP)
- False Negative (FN)



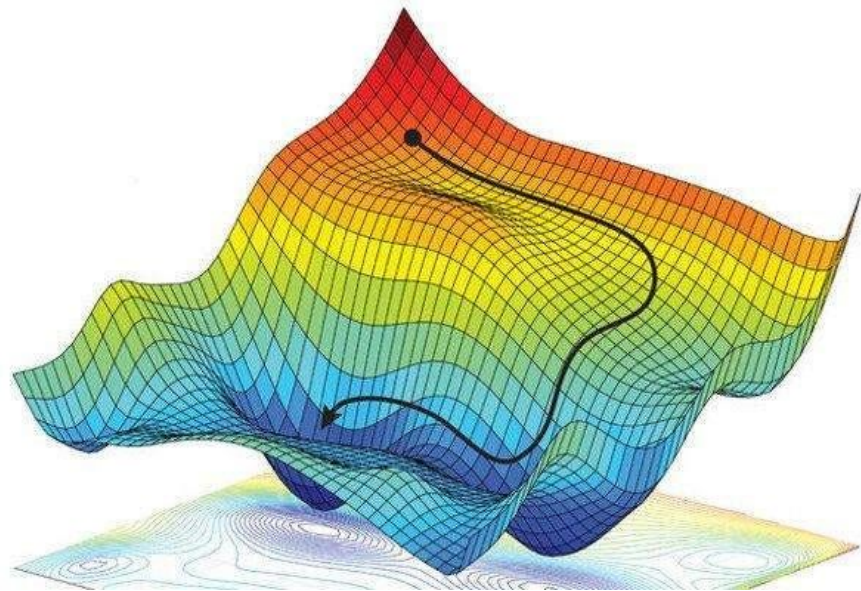
Fine-Tuning A Segmentation Model

- Split the cell sample data into training and testing categories
- Create ground-truth segmentations for baseline comparison



Training Configurations

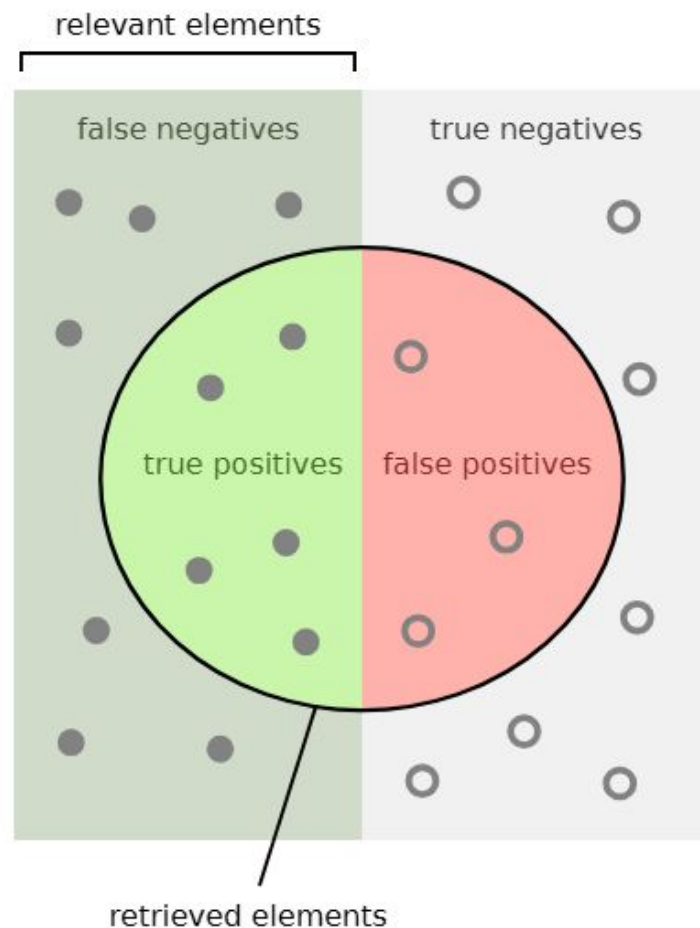
- Epochs: 60 (increments of 10)
 - Complete training cycle
- Learning Rate: 0.01
 - Controls the size of step
- Weight Decay: 0.0001
 - Reduces overfitting



Source: Medium

Metrics Explained

- IOU/Dice Coefficient
 - Segmentation Accuracy
- Precision
 - How conservative is the model?
- Recall
 - How complete was the prediction?

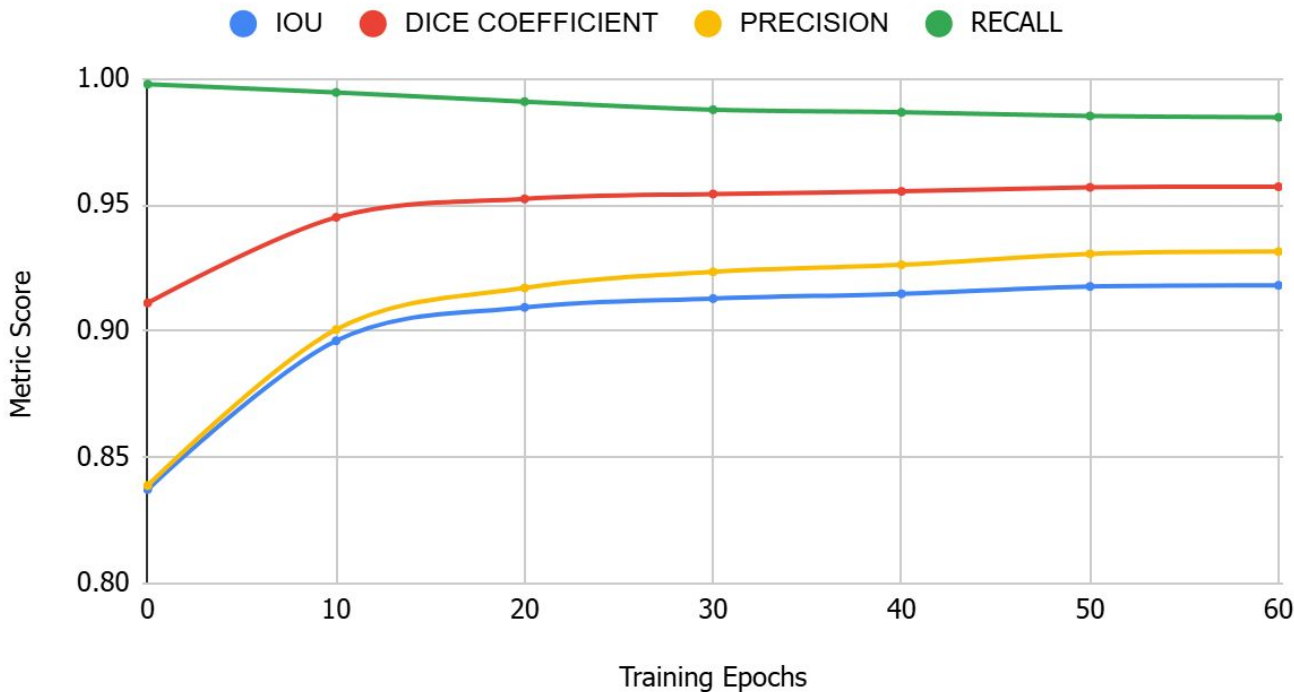


Source: Wikipedia

Fine Tuning Improves Segmentation Performance

- Rapid improvement during early epochs
- Performance stabilizes after 50 epochs

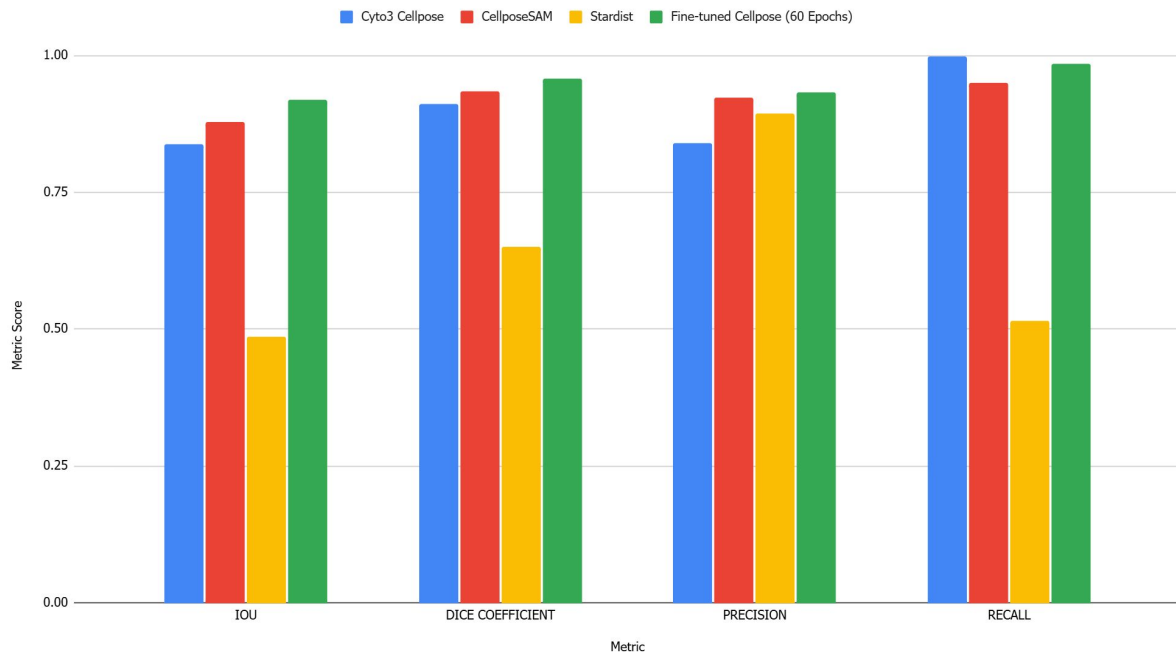
Effect of Training Epochs on Segmentation Performance



Comparative Evaluation Across Segmentation Frameworks

- Fine-tuning improves segmentation quality across multiple metrics
- Cyto3 Cellpose has slightly higher Recall, but with low Precision

Comparison of Segmentation Performance Across Models



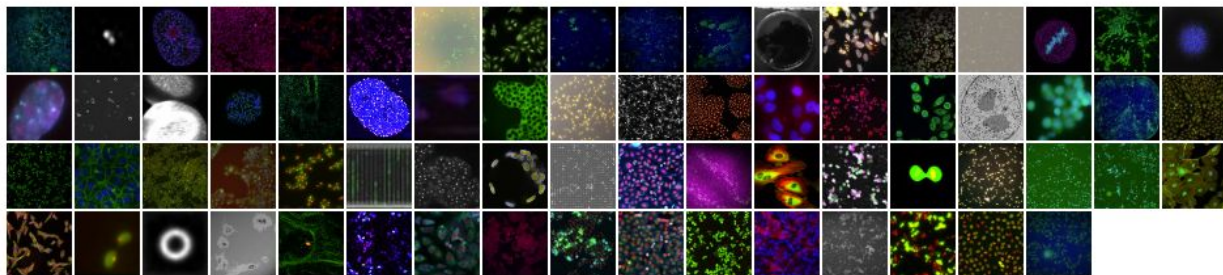
Significance

- Benchmarking framework standardizes segmentation model evaluation on a large scale
- Both model fine-tuning and the benchmarking framework can create efficient workflows

Dataset Name	PC3 Cells Model Data
Dataset Description	
Timepoints	54
XY Slices	1
Z Slices	5
Channels	3
Annotations	3750

Future Directions

- Integrate framework into additional microscopy analysis platforms
- Benchmark segmentation models on variety of datasets
- Incorporate framework into more research workflows



Source: Image Data Resource

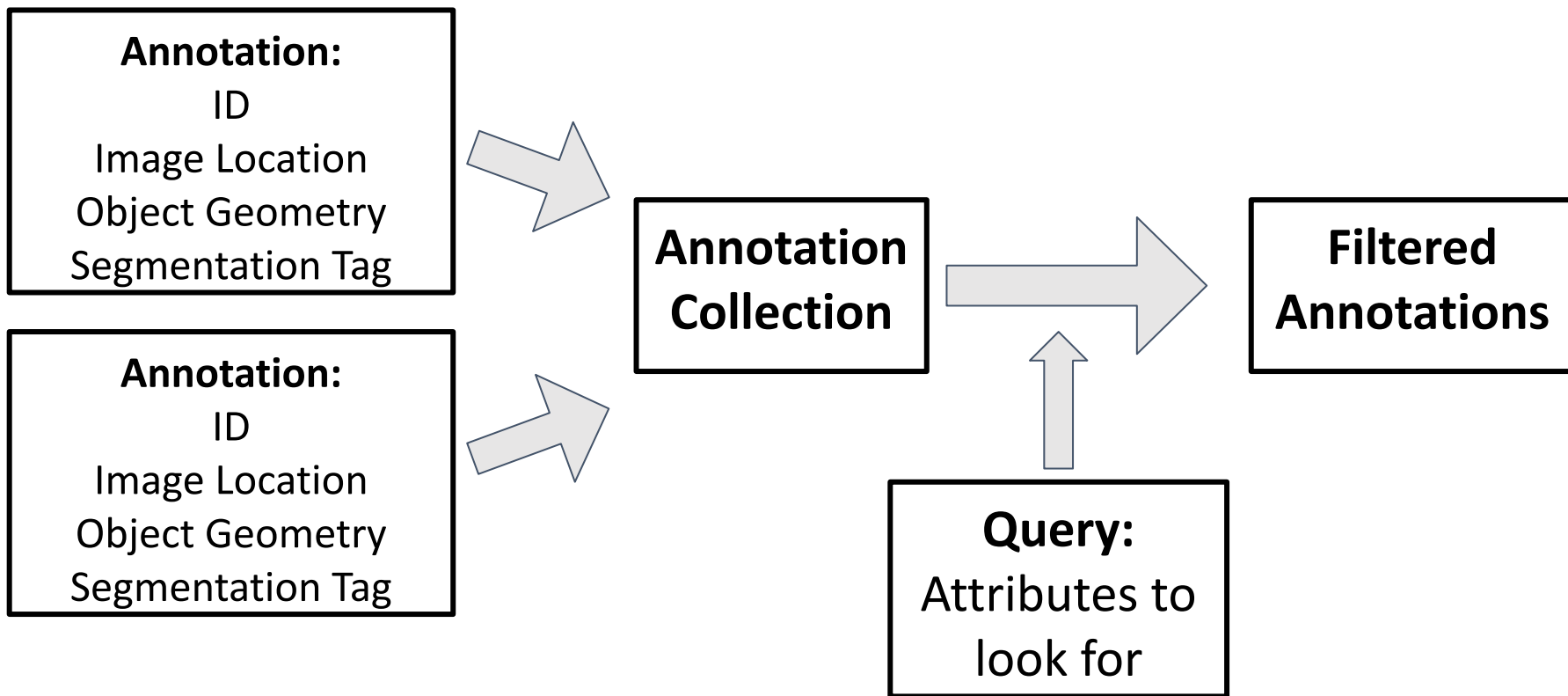
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- Program Coordinators: Laura Rico-Beck and Jessy Morgan
- Organizations: Fei Lab, UChicago, After School Matters

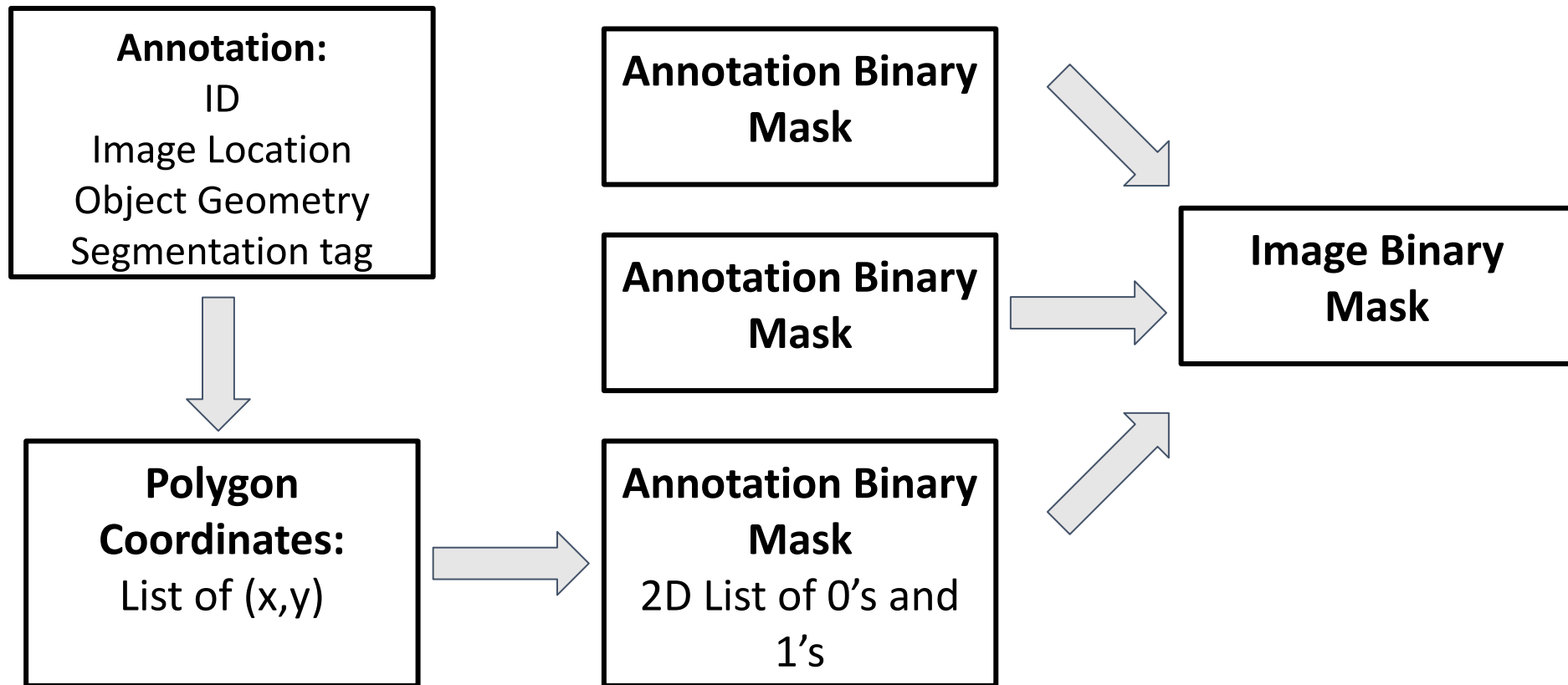


AFTER
SCHOOL matters

Annotation System (Backup)



Binary Masks Explained (Backup)



Training Results in Consistency (Backup)

- Fine-tuned model has the highest Dice coefficient across test images
- Lower variation suggests improved segmentation

